Group Project: Cloud Computing and Ethics

Project: COVID Variant Analysis. We compare the spike protein of Alpha, Delta and Omicron (1,406 sequences, 1,273 positions) and use machine learning to test if a sequence can predict where a sample came from. Data came from NCBI (public).

Part 1: Cloud Computing

1. What is cloud computing and how does it help us?

Cloud computing means renting computers, storage and software over the internet instead of owning them.

Feature	Meaning	Benefit to our project
On-demand self-service	We get resources ourselves, anytime, with no waiting.	We start a notebook when we need it, even at night before a deadline.
Scalability	We can use more power when the work grows.	We can move from 1,406 to 14,965 sequences without buying a new computer.
Elasticity	Resources grow and shrink automatically with demand.	Heavy model runs get more power. Idle times use none.
Resource pooling	Many users share the provider's large pool of machines.	We get strong computers that no student could afford alone.
Pay-as-you-go	We pay only for what we use.	Our dataset is small, so cost is close to zero.

2. Service model: PaaS (Platform as a Service)

Service: Google Colab.

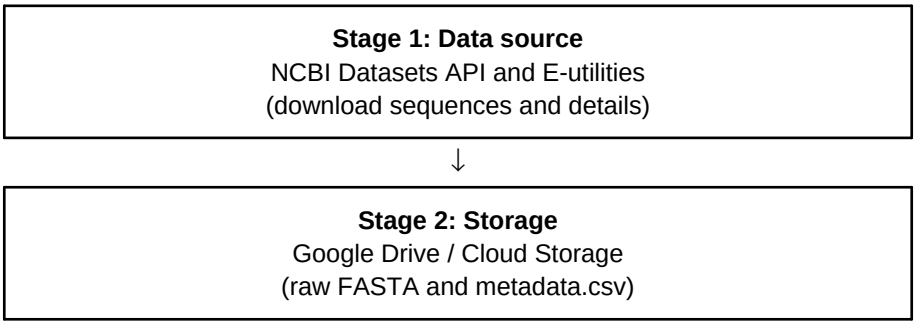
Why PaaS:

- The provider manages the servers and system. We only write and run our code.
- Python, scikit-learn, pandas and Biopython (our tools) are already available or install in one line.
- IaaS (e.g. EC2) would make us set up and maintain the machine ourselves. That wastes time for a small project.
- SaaS gives a finished app, so we could not train our own model.
- Our main file is a notebook, and Colab runs notebooks directly. Group members can share and edit it together.

3. Deployment model: Public cloud

Our data is public and has no personal details. It comes from NCBI (virus sequences and metadata), Nextstrain and CoVariants. It is virus data, not human data, and it has no patient names or IDs. So it needs no private cloud. Public cloud is the cheapest and easiest to share. Hybrid is only needed if we later add private patient data.

4. Cloud pipeline



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Stage 3: Preprocessing
Google Colab (Python: pandas, Biopython)
(remove duplicates, align to Wuhan reference, make 1,406 x 1,273 table of 0/1)

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Stage 4: Model training
Google Colab (scikit-learn, CPU runtime)
(classifier predicts variant and region)

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Stage 5: Results to the user
Notebook + figures + README shared through GitHub
(readers open the link to see the results)

Part 2: Ethics (our project)

1. Ethical issues and how to address them

Issue	Simple explanation	How to address it
Data use and consent	Virus sequences come from real patients. Even without names, the country and date come from them.	Use only public, open data. Follow NCBI's terms. Cite all sources. Never try to identify a person.
Bias in the data (who is missing)	Most samples are from Europe and North America. Africa and other regions are rare, so results may not fit them.	Say clearly that the data is uneven. Use balanced accuracy. Collect more samples from missing regions.
Misleading results	Our first region model looked very accurate (0.944) but was only detecting sub-types.	Test for shortcuts, fix them, and report the lower, honest scores. Do not overclaim.
Misuse of the technology	A region predictor could be used to blame a country or group for the virus.	State that the model is weak (0.66 to 0.94) and is not proof of where a case came from.
Fairness of benefit	Findings should help the places the samples came from.	Share results openly and credit the data providers.

2. Kenya's Data Protection Act 2019: our duties

If we used data from Kenyan patients, genetic and health data are sensitive, so we would have to:

- **Register** with the Office of the Data Protection Commissioner (ODPC) as a data controller or processor.
- **Get clear consent** from patients, or have another legal basis. Tell them why the data is collected and how it will be used.
- **Use the data only for the stated purpose** and collect only what we need (data minimisation).
- **Keep it safe:** remove names and IDs, encrypt it, limit who can open it.
- **Respect patient rights:** to see, correct and delete their data, and to withdraw consent.
- **Do an impact assessment** (DPIA) before starting, because the data is high risk.
- **Report a data breach** to the ODPC within 72 hours, and tell the affected people.
- **Limit transfers abroad.** Sending Kenyan data to a foreign cloud needs safeguards, and sensitive data may need to be kept in Kenya.

- **Keep data only as long as needed,** then delete it.

3. Misuse and harm if wrong

	Answer
Risk	The region model could be wrong or misused. Someone could use it to say a person or group "brought" the virus and treat them unfairly (discrimination, travel bans). A wrong result can also mislead tracing teams.
Safeguard	Always show the limits. Our region accuracy is below the maximum possible (0.871) and the model is only one input beside travel history and contact tracing. Never use it alone to judge a person. Keep the data anonymous.

Part 3: Practice Case, Ziwa Biotech (Gene Drive)

The two issues with the most evidence from the checklist:

Issue	**Evidence in the case**	**How to address it**
1. Ecosystem harm and spread to other species or places	The gene drive passes a sterility gene to almost all offspring and could wipe out a local mosquito population. Mosquitoes are food for fish and other animals. Lake Victoria is connected to other places, so the mosquitoes may spread and the change cannot easily be undone.	Do an independent environmental risk study first. Run small, closed field tests. Make a plan to stop or reverse the drive. Monitor the island after release.
2. Community consent and who decides	Fishing communities learnt of the plan from the news. Nobody asked them.	Talk to the communities before any decision. Explain the risks in simple language and local languages. Let them say no. Involve Kenyan regulators and the lake's neighbouring countries.

Also worth noting: the company announced "the end of malaria in East Africa" before peer review. This is overclaiming. Results should be published and reviewed first.